UNIVERSITÀ DEGLI STUDI DI MILANO

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Text Mining and Sentiment analysis class [NLP]

ANALYZING THEMATIC ALIGNMENT IN SCIENTIFIC JOURNALS

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AI Usage Disclaimer

In preparing this work, I utilized various generative AI tools, primarily ChatGPT and Gemini, for grammar checking, summarizing reference articles, building a comprehensive report, and LaTeX formatting. Additionally, for specific developmental tasks, I used GitHub Copilot for code debugging, and Codex to render the majority of the complex data visualizations. During the methodology development, I used Anthropic's Claude to directly compare HDBSCAN and k-means, as standard web searches did not quickly provide the direct implementation differences I needed. Although the main part of the coding project was based on official documentation, I also utilized Claude to solve a specific problem: integrating objects of different shapes into a single space, since I could not find this specific application in the documentation.

Contents

Chapter 1

Introduction

This project is dedicated to comparing the semantic alignment of articles published within a specific journal with its declared aims and scope (The Lancet Psychiatry, 2026), while also examining the broader range of major topics covered within and beyond its stated mission.

For this purpose, I used an already established NLP framework that quantitatively measures and analyses the semantic alignment between a journal's published articles and its declared Aims and Scope using sentence embeddings, cosine similarity, and BERTopic topic modelling.

The BERTopic pipeline employs a systematic topic reduction mechanism immediately following the initial clustering and representation phases. Once the algorithm has concatenated all documents within each HDBSCAN-derived cluster into a single consolidated document, the c-TF-IDF procedure calculates a distinct topic-word distribution for every cluster. Because density-based clustering can occasionally yield a larger number of highly granular topics than is analytically useful, BERTopic addresses this through an iterative consolidation process. Specifically, the framework compares the c-TF-IDF vectors of the generated topics and iteratively merges the representation of the least common topic with its most semantically similar counterpart.

Chapter 2

Research Question & Methodology

The goal of the project is to see which articles within a journal fit its declared Aims and Scope (The Lancet Psychiatry, 2026), which do not, and what specifically characterizes them. It also aims to investigate whether the number of articles that are closest to the journal's aims and scope outnumbers those that are more distant. Furthermore, the project examines whether document-based semantic similarity is reflected in the similarity of topics identified by BERTopic, and to what extent these topics are aligned with the journal's Aims and Scope.

An additional point of interest is my expectation that the journal will contain predominantly psychiatry-related articles, mostly focusing on academic diseases and highly technical research, while paying less attention to the broader social impact and public health promises stated in the journal's official aims and scope.

For the dataset, I selected PubMed, as it provides a large collection of digitally available and reputable biomedical literature. Within this database, I focused on psychiatry-related publications, although it has a defined disciplinary identity, its boundaries are relatively broad compared with more specialized medical fields. Journals focused on areas such as orthopedics, oncology, or diabetes tend to contain highly specific terminology that directly reflects their domains. In contrast, psychiatric and psychological research often incorporates more interdisciplinary language, resulting in greater semantic overlap across different topics. Which later has been presented as having 17 semantic clusters within only 519 articles, both being similar enough to cluster, and distinctive enough to have so many details within each article.

The choice of psychiatry in contrast of another vague field psychology is in similarity-based search in psychology could therefore produce a higher number of potential false positives due to shared descriptive language across substantially different research areas. Psychiatry stands for middle ground of maintaining clear clinical focus while covering diverse biological, social, and policy-related perspectives. Namely, I selected The Lancet Psychiatry due to its highly selective peer-review process, strong clinical focus, and international reputation in psychiatric research. Additionally, the journal extends beyond clinical treatment by addressing healthcare systems, policy, and social perspectives within mental health, allowing analysis across multiple dimensions of the field.

The foundational Sentence-BERT (SBERT) architecture relies on Siamese and Triplet networks to map text into independent, semantically meaningful dense vector spaces. During training, identical neural networks with shared weights process sentences independently to generate separate embeddings. These embeddings are then evaluated using metrics like contrastive loss or triplet loss, which adjust the network to pull semantically similar texts closer together in the vector space while pushing dissimilar ones apart (Reimers and Gurevych, 2019).

For this implementation, each article was represented by concatenating its title and abstract before generating semantic embeddings using the pre-trained `all-MiniLM-L6-v2` Sentence-BERT model (Hugging Face, 2021). I appreciated its compactness for local machine computing, yet being highly efficient in capturing rich semantic nuances despite its small footprint (1).

Semantic similarity between each article and the journal's mission is then computed using cosine similarity, providing a continuous alignment score that quantifies how closely each publication reflects the editorial objectives of the journal.

Beyond measuring semantic similarity, the project also investigates the latent thematic structure of the publication corpus. Since journals typically publish research across multiple interconnected sub-disciplines, identifying these hidden themes provides additional insight into the relationship between editorial scope and published content. To achieve this, the project employs BERTopic, a transformer-based topic modelling framework that operates through a three-step pipeline. It first generates contextual document embeddings, which are then subjected to dimensionality reduction using Uniform Manifold Approximation and Projection (UMAP) (UMAP Developers, 2026) and density-based clustering through Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN)(McInnes et al., 2026). Finally, it extracts coherent semantic topics from these clusters using a class-based variation of TF-IDF (c-TF-IDF) (Grootendorst, 2022). This combined approach enables the discovery of topics without requiring the number to be specified in advance, while HDBSCAN simultaneously identifies noise documents that do not naturally belong to any major thematic cluster.

The resulting framework provides two complementary perspectives on editorial consistency. First, document-level similarity scores identify publications that are strongly aligned with, or potentially deviate from, the journal's stated objectives. Second, topic modelling reveals how different research themes contribute to the journal's overall scope and whether particular topics exhibit stronger or weaker alignment with the editorial mission. Together, these analyses provide a systematic approach for investigating thematic coherence, identifying potential outlier articles, and leveraging dynamic topic modelling to explore evidence of conceptual evolution within the journal over time.

Chapter 3

Experimental Results

The data were collected automatically using the PubMed Entrez API. All English-language articles published between 2014 (the journal's first year of publication) and May 2026 were collected. Text cleaning was applied by removing unnecessary whitespace, line breaks, and tab characters however no harsh data preprocessing was applied to keep authentic style for sentence embeding. For each article, the title and abstract were concatenated into a single document before being converted into a dense semantic embedding using the Sentence-BERT model all-MiniLM-L6-v2 (1).

My first attempt at understanding the relation and working with semantic vectors was to apply cosine similarity to understand relation between them. Cosine similarity produces values ranging from -1 to 1, where values closer to 1 indicate stronger semantic similarity. In practice, because Sentence-BERT embeddings are contextual semantic representations, the obtained similarity scores are typically positive, with larger values representing articles that are more closely aligned with the journal's intended scope.

The results of the similarity analysis (2) showed that the articles most closely aligned with the aim were those heavily focused on either descriptions of psychiatric diseases or general overviews of psychiatry (1). In contrast, the least preferred articles were those leaning heavily toward socio-economic impact and broad, interdisciplinary, human-focused discussions expressed in vague terms and not directly related to the stated aim. Although the aim and scope mention the importance of highlighting such work, relying on similarity alone would not fully satisfy the long-term objectives of this study.

The BERTopic model identified 17 interpretable topics, each represented by the most characteristic c-TF-IDF keywords. Rather than predefining the number of topics, HDBSCAN determined the natural grouping structure directly from the embedding space (3). The hierarchical clustering(4) demonstrates that the discovered topics form a continuous spectrum of psychiatric research rather than isolated categories. Closely related clinical themes such as public mental health, mood disorders, and severe psychiatric conditions (e.g., schizophrenia and trauma) merge early in the dendrogram, suggesting substantial semantic overlap. In contrast, highly specialized subjects like substance use disorders, ADHD, and neuroimmunology connect much later in the hierarchy, indicating that they maintain distinct vocabularies while still belonging to the broader psychiatric domain. This structural continuum is further supported by the topic similarity matrix(5), where cosine similarities generally range from 0.5 to 0.8. The heatmap reveals gradual transitions between related topics, with the strongest relationships appearing among depression, bipolar disorder, and trauma research. Importantly, the absence of uniformly high similarity blocks indicates that BERTopic successfully separated overlapping subfields without producing redundant clusters(6). Ultimately, these moderate-to-high similarity scores reflect the natural thematic homogeneity expected from a highly specialized psychiatry journal(7).

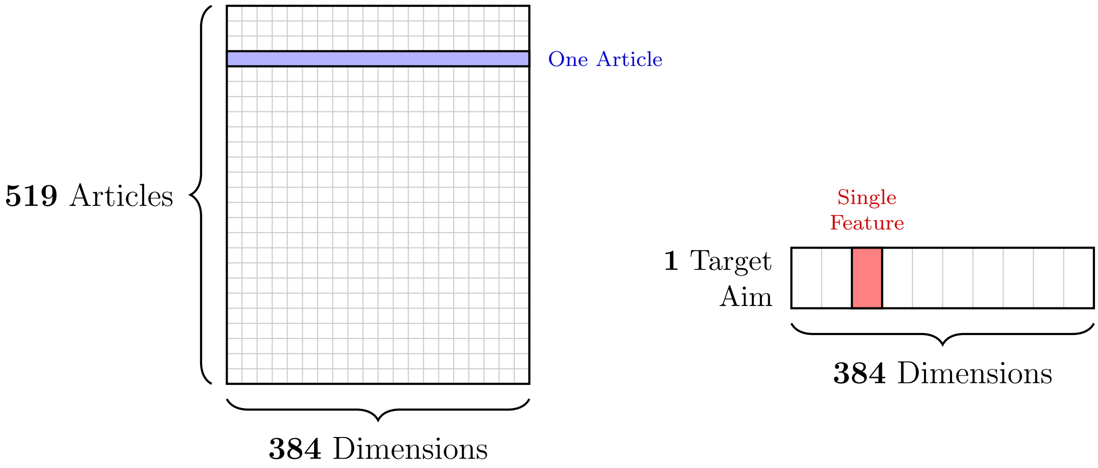
Chapter 4

Concluding Remarks

To extend the work next time, I want to choose a more extensive dataset to actually stress-test the SBERT 5 seconds vs. BERT 65 hours metric. Right now, even with the comprehensive list of clusters made due to the large and cross-disciplinary nature of the topic, next time I want to do more interesting work by also expanding the dataset. I could also compare psychiatry vs. psychology journals, instead of just the aims and scope, to see how much crossover they have within their articles and to actually understand what makes an article get published under an umbrella term.

Here, I attempted to compare cosine similarity against BERTopic alignment with the aim, which showed how close the articles were. However, I did it wrong in the sense that the cosine similarities were not clustered at the end; I just evaluated them individually. Next time, I am interested in comparing cosine similarity + PCA vs. BERTopic.

Appendix: Figures and Visualizations



(a) Journal Corpus Matrix: $(519, 384)$

(b) Aim Scope Vector: $(384,)$

Figure 1: Embedding dimensions for Journal Matrix and Target Aim.

Table 1: Top Articles by Semantic Similarity to the Journal's Aims and Scope

Article Order	Title	Similarity Score
518	detection and treatment of at-risk mental state for developing a first psychosis: making up the balance	0.60686636
476	drug development in psychiatry: 50 years of failure and how to resuscitate it	0.587713
376	clinical and social factors associated with increased risk for involuntary psychiatric hospitalisation: a systematic review,...	0.5805417
235	perspectives on icd-11 to understand and improve mental health diagnosis using expertise by experience (include study): an...	0.56905854

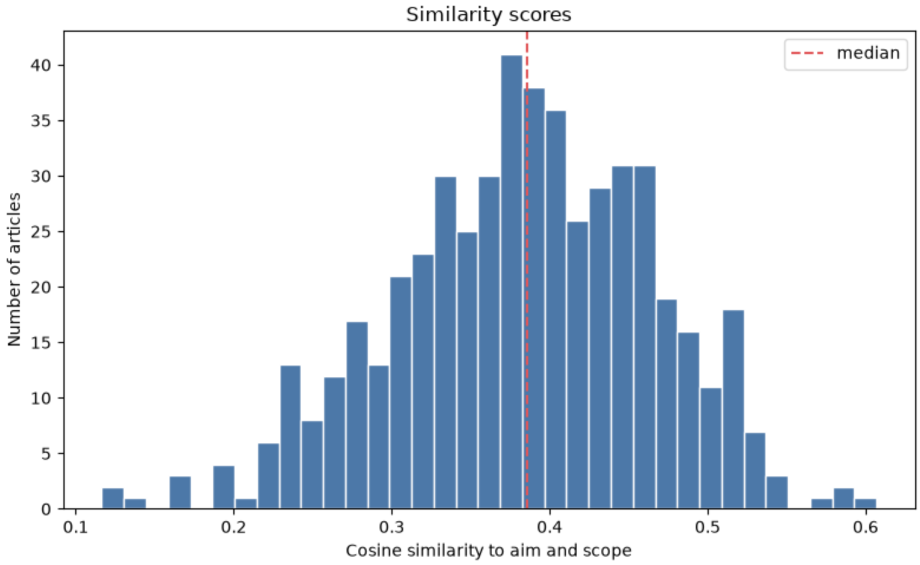


Figure 2: Articles similarity to aim diagram.

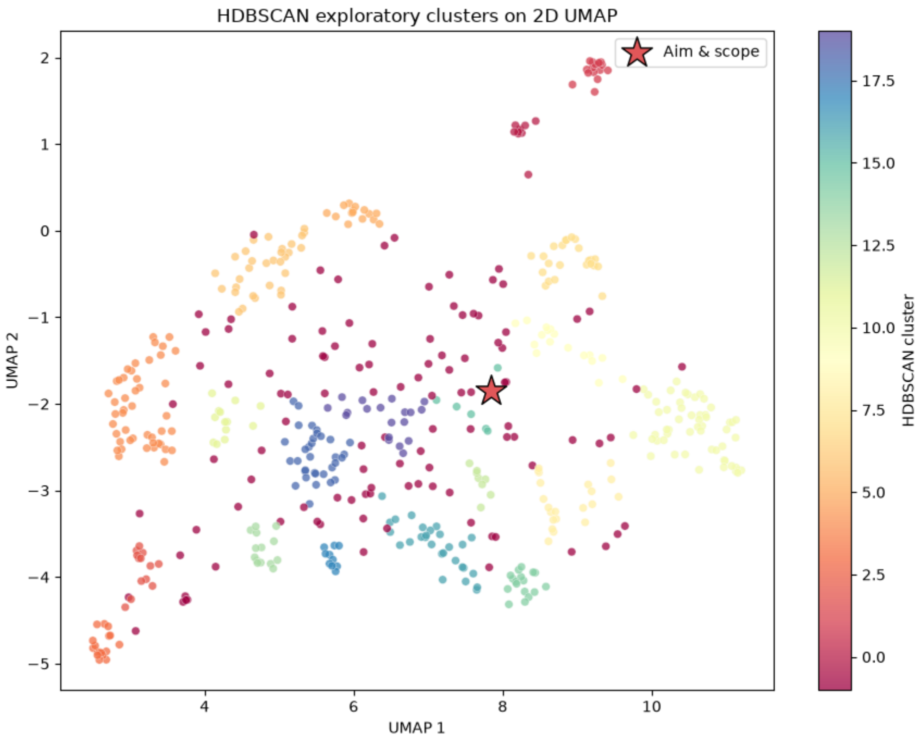


Figure 3: HDBSCAN exploratory clusters on 2D UMAP.

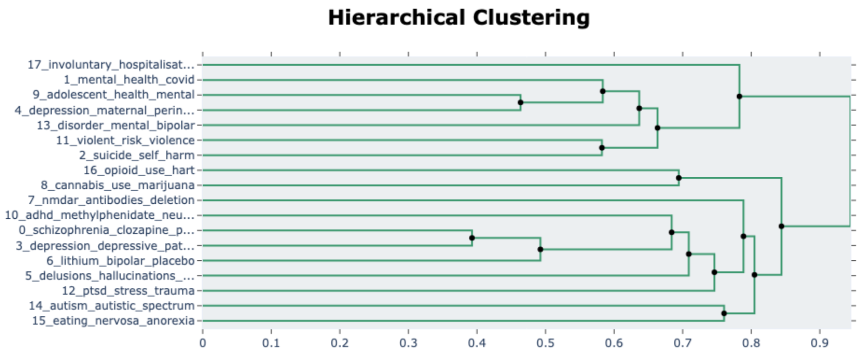


Figure 4: Hierarchical clustering of BERTopic topics.

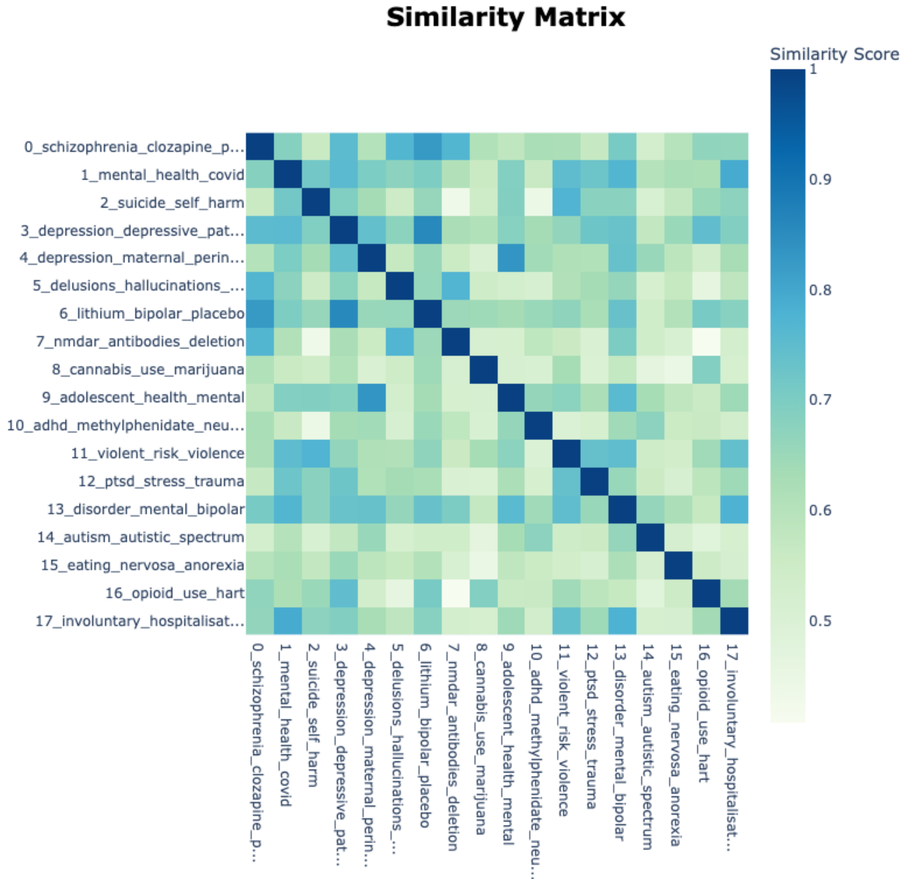


Figure 5: Topic similarity matrix heat map.

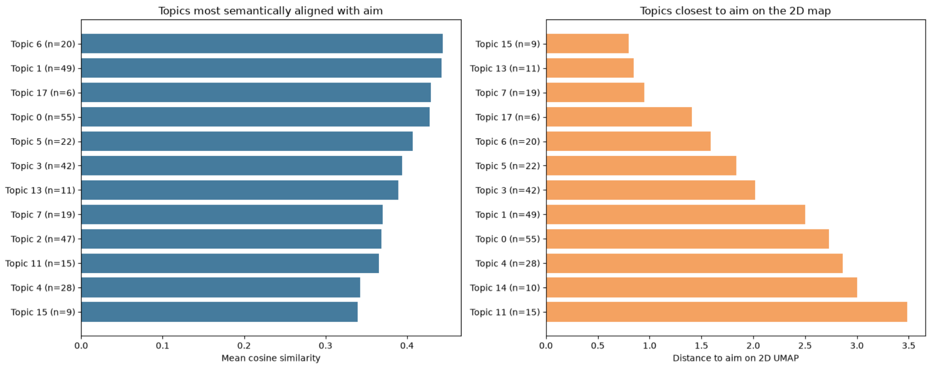


Figure 6: Cosine similarity analysis and final clustering view 1.

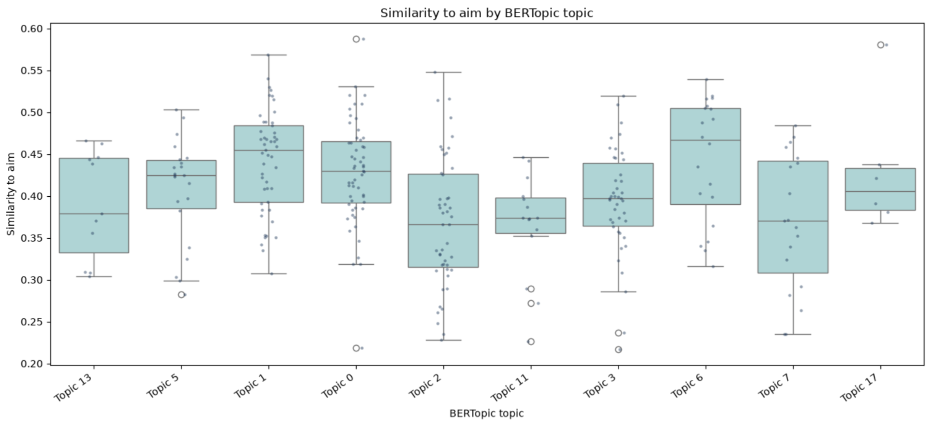


Figure 7: Cosine similarity analysis and final clustering view 2.

Bibliography

Grootendorst, M. (2022). Bertopic: Neural topic modeling with a class-based tf-idf procedure.

Hugging Face (2021). Sentence-transformers: all-minilm-l6-v2 model documentation. Pre-trained model repository.

McInnes, L., Healy, J., and Astels, S. (2026). How HDBSCAN works — hdbscan documentation. `https://hdbscan.readthedocs.io/en/latest/how_hdbscan_works.html`.

Reimers, N. and Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In Inui, K., Jiang, J., Ng, V., and Wan, X., editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, Hong Kong, China. Association for Computational Linguistics.

The Lancet Psychiatry (2026). Aims and scope. Publisher: Elsevier. Accessed online.

UMAP Developers (2026). Comparison of dimension reduction techniques — umap documentation. `https://umap-learn.readthedocs.io/en/latest/auto_examples/plot_algorithm_comparison.html`.